Locally Invertible, Globally Not

The 2026 counterexample to the Jacobian conjecture,
and why the plane is still open

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Exposition only — all mathematics is due to the cited authors.

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Abstract

In 1939 Keller asked whether a polynomial map of $\mathbb{C}^n$ to itself whose Jacobian determinant is a nonzero constant must have a polynomial inverse. In July 2026 Levent Alpöge announced an explicit map of $\mathbb{C}^3$ with Jacobian determinant -2 that sends three different points to the same point. So the answer is no in dimension three, and hence in every dimension from three up. Dimension two, the case MathDB lists as JSP-000037, is still open. This guide is written for a reader who knows calculus and linear algebra. It checks the counterexample by hand, explains why it works (a cubic polynomial can be split into a linear factor times a quadratic factor in three different ways), and shows why the map has no inverse. It then describes what the formal proofs certify, including the author's Lean refutation of MathDB's formal statement, which covers every dimension, and explains why the two-dimensional case is untouched. Citations [Key, place] name a source from the bibliography and the place in it; [background: ...] notes give context from the wider literature.

Where things stand

- **Proved by direct computation** [Tao, eq. (1)]: an explicit polynomial map $F : \mathbb{C}^3 \to \mathbb{C}^3$ has $\det DF = -2$ everywhere, yet $F(0, 0, -\frac{1}{4}) = F(1, -\frac{3}{2}, \frac{13}{2}) = F(-1, \frac{3}{2}, \frac{13}{2})$. Hence the Jacobian conjecture is false in every dimension $n \geq 3$.
- **Formally checked**: in Isabelle/HOL [AFP] and in Lean 4. The author's Lean proof [XL] shows that MathDB's formal statement for this problem, which claims every dimension $n \geq 1$, is false.
- **Not refereed**: the announcement and the 2026 follow-up papers. The counterexample itself is a finite computation that anyone can redo.
- **Still open**: dimension two, the plane case. It is the problem as MathDB states it [Off].

1 The question

A *polynomial map* $F = (F_1, \dots, F_n) : \mathbb{C}^n \to \mathbb{C}^n$ is a list of n polynomials in n variables. Its *Jacobian matrix* DF is the $n \times n$ matrix of partial derivatives $\partial F_i / \partial z_j$, and its *Jacobian determinant* $\det DF$ is again a polynomial. *F has a polynomial inverse* if some polynomial map G satisfies $G(F(z)) = z$ and $F(G(z)) = z$ for all z.

If F has a polynomial inverse G, the chain rule gives $DG(F(z))\, DF(z) = I$, so $\det DG(F(z)) \cdot \det DF(z) = 1$. The product of two polynomials can be the constant 1 only if both are nonzero constants. So an invertible polynomial map always has a nonzero constant Jacobian determinant [vdE, §2, p. 2]. Keller asked whether the converse holds [Kel]:

Conjecture (Jacobian conjecture, JC_n)**.** *If $F : \mathbb{C}^n \to \mathbb{C}^n$ is a polynomial map and $\det DF$ is a nonzero constant, then F has a polynomial inverse.*

Maps with $\det DF$ a nonzero constant are called *Keller maps*. Three remarks set the stage.

- **One variable is easy.** If $F'(z)$ is a nonzero constant a, then $F(z) = az + b$, whose inverse is $(w - b)/a$ [MY, Thm. 4.1(i), p. 4].
- **Only injectivity matters.** Białynicki-Birula and Rosenlicht proved that a polynomial map $\mathbb{C}^n \to \mathbb{C}^n$ that is one-to-one is automatically onto, with a polynomial inverse [BBR]. So a counterexample is exactly a Keller map that sends two different points to the same point [Gao, §2, p. 3].
- **The field does not matter.** Over any field of characteristic zero the question is the same, and it suffices to settle it over $\mathbb{C}$. MathDB states it for every field of characteristic zero and notes, citing [BCW], that the complex case suffices [Off].

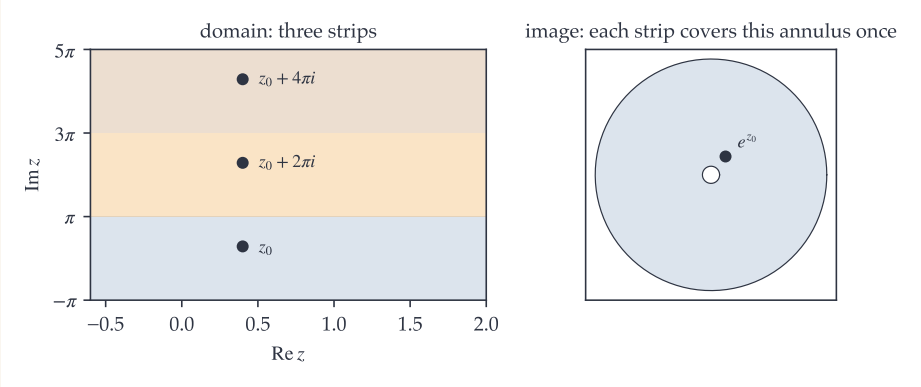
MathDB's problem page states only the plane case $n = 2$, notes that the general form is false for every $n \geq 3$, and records that "only $n = 2$ remains open" [Off]. The prize catalog entry for JSP-000037 covers "both dimension two and dimensions $n \geq 3$" [JSP]. This guide treats both: the counterexample for $n \geq 3$ and the open plane case.

2 Local versus global

By the inverse function theorem, if $\det DF(z_0) \neq 0$ then F is one-to-one on a small ball around z_0, with a smooth inverse there. A Keller map is therefore invertible *near every point*. The conjecture asks whether invertible near every point means invertible everywhere [Tao, after Conj. 1].

For a Keller map, two preimages can never merge over a target point: near each preimage the map is a local change of coordinates. So the only way to lose injectivity is for preimages to "escape to infinity" as the target point moves [Gao, §2, p. 3]. The question is whether a polynomial map can have more than one sheet in this way. In 2026 the answer turned out to be: yes, three.

Picture, not proof



For general smooth or analytic maps the answer is plainly no. The complex exponential $z \mapsto e^z$ has derivative $e^z \neq 0$ everywhere, so it is invertible near every point. But $e^{z+2\pi i} = e^z$: each horizontal strip of height 2π is wrapped once around the same annulus, and the points $z_0,\ z_0 + 2\pi i,\ z_0 + 4\pi i, \ldots$ all land on one point. The exponential is not a polynomial, and the hope behind the conjecture was that polynomials are too rigid to wrap around like this.

3 The counterexample

On July 19, 2026, Levent Alpöge announced the following map [Alp]. The announcement is not a refereed paper; its content is a finite computation. It credits Akhil Mathew with prompting the question and an AI system with work leading to the map [AFP, abstract].

$$\begin{aligned} F_1 &= (1 + z_1 z_2)^3 z_3 + z_2^2 (1 + z_1 z_2)(4 + 3 z_1 z_2), \\ F_2 &= z_2 + 3 z_1 (1 + z_1 z_2)^2 z_3 + 3 z_1 z_2^2 (4 + 3 z_1 z_2), \\ F_3 &= 2 z_1 - 3 z_1^2 z_2 - z_1^3 z_3. \end{aligned} \tag{1}$$

Theorem ([Tao, eq. (1)]; [Gao, Thm. 3.3, p. 6]). *The map* $F = (F_1, F_2, F_3) : \mathbb{C}^3 \to \mathbb{C}^3$ *has* $\det DF = -2$ *at every point, and*

$$F(0, 0, -\tfrac{1}{4}) \;=\; F(1, -\tfrac{3}{2}, \tfrac{13}{2}) \;=\; F(-1, \tfrac{3}{2}, \tfrac{13}{2}) \;=\; (-\tfrac{1}{4}, 0, 0).$$

The components have degrees 7, 6 and 4 [Gao, Thm. 3.3, p. 6]. A priori $\det DF$ is a polynomial of degree up to 18; all its non-constant terms cancel [Tao, after eq. (1)]. The determinant is a routine expansion of a 3×3 determinant; the collision can be checked by hand.

Three points, one image: a hand check

At $(0,0,-\frac{1}{4})$**.** Every term containing z_1 or z_2 vanishes, so $F = (1 \cdot z_3,\ 0,\ 0) = (-\frac{1}{4}, 0, 0)$.

At $(1,-\frac{3}{2},\frac{13}{2})$**.** Here $1 + z_1 z_2 = -\frac{1}{2}$ and $4 + 3z_1 z_2 = -\frac{1}{2}$. Then

$$\begin{aligned} F_1 &= (-\tfrac{1}{2})^3 \cdot \tfrac{13}{2} + \tfrac{9}{4} \cdot (-\tfrac{1}{2})(-\tfrac{1}{2}) = -\tfrac{13}{16} + \tfrac{9}{16} = -\tfrac{1}{4}, \\ F_2 &= -\tfrac{3}{2} + 3 \cdot \tfrac{1}{4} \cdot \tfrac{13}{2} + 3 \cdot \tfrac{9}{4} \cdot (-\tfrac{1}{2}) = -\tfrac{12}{8} + \tfrac{39}{8} - \tfrac{27}{8} = 0, \\ F_3 &= 2 + \tfrac{9}{2} - \tfrac{13}{2} = 0. \end{aligned}$$

At $(-1,\frac{3}{2},\frac{13}{2})$**.** Replacing (z_1, z_2) by $(-z_1, -z_2)$ keeps F_1 and flips the signs of F_2 and F_3. Since $F_2 = F_3 = 0$ at the previous point, the value is again $(-\frac{1}{4}, 0, 0)$.

So F is invertible near every point but sends three points to one: the Jacobian conjecture is false in dimension three. By the theorem of Białynicki-Birula and Rosenlicht, failing injectivity is the only way a Keller map can fail; Section 5 shows directly that F has no polynomial inverse.

4 Why it works: a line times a quadric

Written as (1), the map looks like a miracle. Tao's account [Tao] explains where the three preimages come from, using nothing deeper than multiplying polynomials.

Multiplying forms. Take a *linear form* $L(z,w) = az + bw$ and a *quadratic form* $Q(z,w) = cz^2 + dzw + ew^2$ in two auxiliary variables z, w. Their product is a cubic form

$$LQ = f\,z^3 + g\,z^2 w + h\,z w^2 + i\,w^3, \qquad (f,g,h,i) = (ac,\ ad+bc,\ ae+bd,\ be)$$

[Tao, eq. (2)]. A cubic form with three distinct linear factors, $C = L_1 L_2 L_3$, can be written as (linear)×(quadratic) in three ways: $L_1 \cdot (L_2 L_3)$, $L_2 \cdot (L_1 L_3)$ and $L_3 \cdot (L_1 L_2)$ [Tao, eq. (4)]. Multiplication is therefore three-to-one, once one discards the trivial ambiguity $(L, Q) \mapsto (\lambda L, \lambda^{-1} Q)$.

Removing the trivial ambiguity. The *resultant*

$$\operatorname{Res}(L,Q) = \det \begin{pmatrix} a & b & 0 \\ 0 & a & b \\ c & d & e \end{pmatrix} = a^2 e - abd + cb^2$$

vanishes exactly when L and Q share a root, and the rescaling $(\lambda L, \lambda^{-1} Q)$ multiplies it by λ [Tao, eqs. (5)–(6)]. So, as long as L and Q share no root, imposing $\operatorname{Res}(L,Q) = 1$ picks exactly one representative of each splitting [Tao, eq. (7)]. Finally Tao restricts to cubics whose $z^2 w$-coefficient is $g = 1$.

The dictionary. These two conditions cut out a three-dimensional set of pairs (L, Q), and Tao shows that it is a copy of $\mathbb{C}^3$ with polynomial coordinates (a, y, z), where $b = 1 + ay$ [Tao, eqs. (10)–(16) and after]. Rewritten in the variables of (1), the dictionary reads: to the point (z_1, z_2, z_3) attach

$$\begin{aligned} L &= z_1\, z + (1 + z_1 z_2)\, w, \\ Q &= \big(1 - \tfrac{3}{2} z_1 z_2 - \tfrac{1}{2} z_1^2 z_3\big) z^2 + \big(\tfrac{1}{2} z_2 + \tfrac{1}{2} z_1 z_3 + \tfrac{3}{2} z_1 z_2^2 + \tfrac{1}{2} z_1^2 z_2 z_3\big) z w \\ &\quad + \big(z_3 + 4 z_2^2 + 2 z_1 z_2 z_3 + 3 z_1 z_2^3 + z_1^2 z_2^2 z_3\big) w^2. \end{aligned}$$

Then $\mathrm{Res}(L, Q) = 1$ and $g = 1$ identically, and

$$F(z_1, z_2, z_3) = (i,\ 2h,\ 2f),$$

the remaining coefficients of the cubic LQ. This is Tao's final identity [Tao, last display] in the coordinates $(z_1, z_2, z_3) = (a, y, -2z)$, where it reads $F(a, y, -2z) = (G_3, 2G_2, 2G_1)$; all three identities can be confirmed by expanding. In words: *F multiplies a linear form by a quadratic form and reads off the coefficients of the product.*

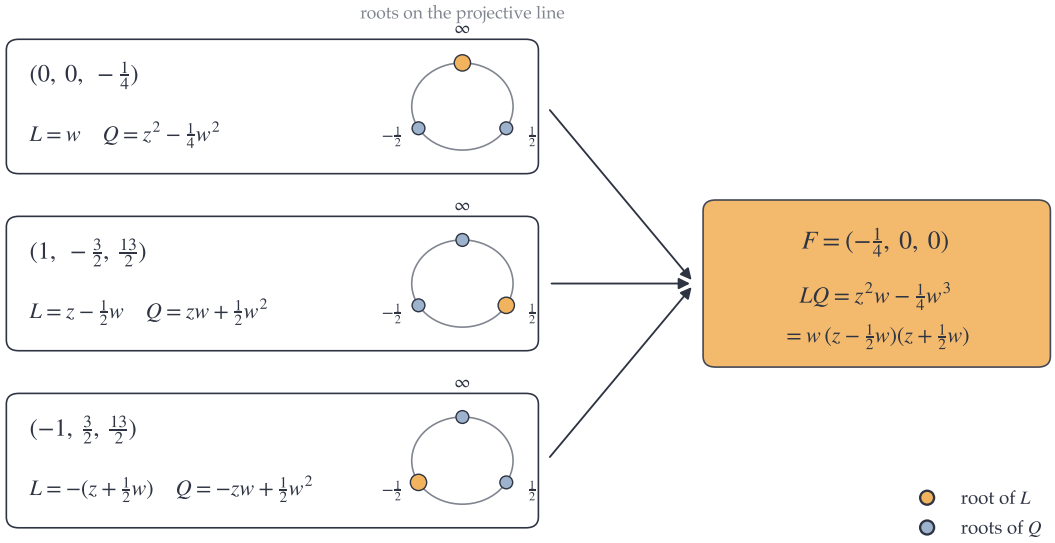


Figure 1: The three preimages of $(-\frac{1}{4}, 0, 0)$ are the three ways to split the cubic $z^2w - \frac{1}{4}w^3 = w\,(z - \frac{1}{2}w)(z + \frac{1}{2}w)$ as a linear form L times a quadratic form Q with $\mathrm{Res}(L, Q) = 1$. On each circle (the roots z/w on the projective line) the amber dot is the root given to L.

The three points, explained. The common value $(-\frac{1}{4}, 0, 0)$ means $f = h = 0$, $g = 1$, $i = -\frac{1}{4}$: the cubic $z^2w - \frac{1}{4}w^3 = w\,(z - \frac{1}{2}w)(z + \frac{1}{2}w)$. Its three splittings with Res $= 1$ are (Figure 1)

$$\left(w,\ z^2 - \tfrac{1}{4}w^2\right), \qquad \left(z - \tfrac{1}{2}w,\ zw + \tfrac{1}{2}w^2\right), \qquad \left(-(z + \tfrac{1}{2}w),\ -zw + \tfrac{1}{2}w^2\right),$$

and the dictionary sends them back to $(0, 0, -\frac{1}{4})$, $(1, -\frac{3}{2}, \frac{13}{2})$ and $(-1, \frac{3}{2}, \frac{13}{2})$. The first coordinate $z_1 = a$ is simply the z-coefficient of L. A generic cubic with $g = 1$ has three distinct linear factors, and accordingly a generic point has exactly three preimages [Gao, Thm. 3.3, p. 6].

Picture, not proof: why the determinant is constant

Near a pair (L, Q) with $\mathrm{Res}(L, Q) \neq 0$, the root of L is apart from the roots of Q. Moving the cubic $C = LQ$ a little moves its three roots a little, and the split "this root to L, the other two to Q" persists. So nearby cubics have exactly one nearby splitting: the map is invertible near every point [Tao, after eq. (8)]. A polynomial on $\mathbb{C}^3$ that never vanishes must be constant, so local invertibility everywhere forces $\det DF$ to be a nonzero constant.

5 Why there is no polynomial inverse

Two short arguments, either of which suffices.

The collision. If G were a polynomial map with $G(F(z)) = z$ for all z, then $(0,0,-\frac{1}{4}) = G(F(0,0,-\frac{1}{4})) = G(F(1,-\frac{3}{2},\frac{13}{2})) = (1,-\frac{3}{2},\frac{13}{2})$, which is false. Isabelle proves that F has no polynomial inverse [AFP, p. 10]; the author's Lean proof [XL] uses exactly this argument.

A component that factors. Van den Essen points out an even shorter reason, which needs no evaluation at all [vdE, §2, p. 2]. The third component factors:

$$F_3 = 2z_1 - 3z_1^2 z_2 - z_1^3 z_3 = -z_1\left(z_1^2 z_3 + 3z_1 z_2 - 2\right).$$

A component of an invertible map cannot factor. Indeed, if $F(G(z)) = z$ then $F_3(G(z)) = z_3$. Writing $F_3 = uv$ with u, v non-constant gives $u(G(z)) \cdot v(G(z)) = z_3$. Since z_3 cannot be written as a product of two non-constant polynomials, one factor, say $u(G(z))$, is a constant. But G is onto (it has the inverse F), so u itself would be constant, a contradiction. Van den Essen credits this observation, that F_3 is a "fake coordinate", to Stefan Maubach [vdE, §2, p. 2].

Not even onto. Gao determines all fibers of F by exact computation: every point has 3, 1 or 0 preimages, never 2, and the image of F is $\mathbb{C}^3$ minus the curve $\{(\frac{4}{27}t^{-2}, \frac{4}{3}t^{-1}, t) : t \neq 0\}$ [Gao, Thm. 3.4, pp. 6–7]. Gao's paper is a preprint. So two of the three sheets can "escape to infinity" together, and over the missing curve all three do.

6 Every dimension from three up

For $n \geq 4$, pad F with identity coordinates:

$$\tilde{F}(z_1, \ldots, z_n) = \big(F_1(z_1, z_2, z_3),\, F_2(z_1, z_2, z_3),\, F_3(z_1, z_2, z_3),\, z_4, \ldots, z_n\big).$$

Its Jacobian matrix is block diagonal, with DF in one block and the identity in the other, so $\det D\tilde{F} = -2$. The three colliding points, padded with the same extra coordinates, still collide. Hence JC_n is false for every $n \geq 3$ [AFP, p. 18]; [MY, Thm. 4.1(ii), p. 4].

The example is not isolated. Within days, Gallagher gave a construction producing, for every $m \geq 3$, a polynomial map of $\mathbb{C}^3$ with Jacobian determinant 1 whose generic fiber has exactly m points [Gal]. Speyer explained the geometry behind it, the *tangent sweep* [SpeN, pp. 1–2], and Gao gave a systematic version with fully worked examples, including one of generic fiber size four [Gao, Thm. 3.5, p. 7]. Van den Essen showed how a counterexample can be found by elementary means [vdE, §1]. All of these are preprints or notes.

7 What the formal proofs certify

7.1 Isabelle/HOL

An entry of the Archive of Formal Proofs by Freitas Ramos, Barros Hulak and de Queiroz verifies the map (1) [AFP]. It proves that the determinant is identically -2 [AFP, p. 8], that the three points

collide, that the map has no polynomial inverse [AFP, p. 10], and that padding gives counterexamples in every dimension $n \geq 3$ [AFP, p. 18]. The entry states that it "makes no claim about the two-dimensional conjecture" [AFP, abstract]. Independent Lean 4 checks of the determinant and the collision appeared within days [SM, NG].

7.2 MathDB's formal statement

MathDB attaches the following Lean statement to the problem [Tgt] (Unicode symbols typeset):

```
abbrev Target : Prop :=
  ∀ (K : Type u) [Field K] [CharZero K] (n : ℕ), 1 ≤ n →
  ∀ F : Fin n → MvPolynomial (Fin n) K,
    (∃ c : K, c ≠ 0 ∧
       Matrix.det (fun i j => MvPolynomial.pderiv j (F i)) = MvPolynomial.C c) →
    ∃ G : Fin n → MvPolynomial (Fin n) K,
      (∀ i, MvPolynomial.bind₁ G (F i) = MvPolynomial.X i) ∧
      (∀ i, MvPolynomial.bind₁ F (G i) = MvPolynomial.X i)
```

Here `bind₁ G (F i)` substitutes the polynomials $G_1, \ldots, G_n$ into F_i, that is, it is $F_i(G)$. Comparing clause by clause with the official text [Off, JSP]:

No.	Official text	Lean	Reading
(1)	every field K of characteristic 0	`[Field K] [CharZero K]`	same
(2)	MathDB: $n = 2$ only [Off]; prize catalog: $n = 2$ and $n \geq 3$ [JSP]	`(n : ℕ), 1 ≤ n`	every dimension, as in Keller's general question
(3)	polynomials $p, q \in K[x, y]$	`F : Fin n → MvPolynomial (Fin n) K`	n polynomials in n variables
(4)	$J_F = c, c \neq 0$	$\det(\partial F_i/\partial x_j)$ is the constant `c`, with `c` $\neq 0$	same
(5)	$g(p,q) = x, h(p,q) = y$, $p(g,h) = x, q(g,h) = y$	$F_i(G) = x_i$ and $G_i(F) = x_i$ for all i	same: a two-sided polynomial inverse

Clauses (1), (3), (4) and (5) match. Clause (2) decides the matter: the formal statement quantifies over every dimension $n \geq 1$, so it includes $n = 3$, and the map (1) shows it is *false*. No proof of it can exist. A formal statement of the plane case alone would fix $n = 2$.

7.3 A Lean refutation of the formal statement

The author's Lean 4 repository [XL] imports MathDB's statement unchanged and proves

```
theorem Submission.solution : ¬ Problem.Target
```

The proof takes $K = \mathbb{Q}$, $n = 3$, $c = -2$ and the map (1). It proves the determinant identity symbolically, evaluates F at $(0, 0, -\frac{1}{4})$ and $(1, -\frac{3}{2}, \frac{13}{2})$, and derives a contradiction from any G with $G_i(F) = x_i$, exactly as in Section 5. Lean's kernel checks it using only the standard axioms `propext`, `Classical.choice` and `Quot.sound`, with no `sorry`. The mathematics is Alpöge's; the formalization is independent.

7.4 What the formal proofs do not certify

They certify the formal statements and nothing else. In particular none of them says anything about $n = 2$: the refutation above only shows that the general-dimension statement is false.

8 Why the plane is still open

What is known in two variables.

- **Low degree.** Wang proved the conjecture for maps of degree at most 2 in every dimension [Wang]; the counterexample (1) has degree 7 [Gao, §2, p. 3].
- **Degree up to 100.** In the plane, Moh verified the conjecture when both components have degree at most 100 [Moh], [Gao, §2, p. 3].
- **Prime degree.** In the plane, if one component of a Keller map has prime degree, the map is invertible [Mag], [vdE, §2, p. 2].

The classical reduction does not help. Bass, Connell and Wright, and Yagzhev, showed that it is enough to prove the conjecture for maps $x + H(x)$ with H homogeneous cubic, *in all dimensions at once*; Drużkowski reduced further to cubes of linear forms [BCW, Yag, Dru]. These reductions raise the dimension, so they say nothing about a fixed dimension such as $n = 2$ [Gao, §2, p. 3].

Why the 2026 ideas use a third variable. Alpöge's map and Gallagher's family both come from a construction in the plane, Speyer's *tangent sweep* [Gao, §3]. The sweep easily produces maps of the plane whose Jacobian determinant is a polynomial of degree one (not a constant), in any degree. Gallagher's further tricks lower the Jacobian's degree from 1 to 0 "at the expense of moving from $\mathbb{A}^2$ to $\mathbb{A}^3$" [SpeN, p. 1]. Van den Essen's search meets a related wall: the simplest two-variable candidate with a factoring component, $f = x + x^2y$, has degree 3, a prime, so by Magnus's theorem it cannot be completed to a two-variable counterexample [vdE, §2, pp. 2–3]. None of this is a proof for the plane; it only explains why the 2026 ideas do not reach it.

Where the plane case sits. Meng and Yang note that the Hessian conjecture in four variables implies JC_2, and that among the statements JC_n and HC_n ($n \geq 1$) these two are now the only open cases [MY, Thm. 4.1(iv), p. 4] (a preprint). Over the reals, a plane map whose Jacobian determinant never vanishes, but is not constant, need not be injective (Pinchuk's counterexample [Pin]); the constant Jacobian case over $\mathbb{C}$ is the one that remains. The problem was on Smale's list of problems for the twenty-first century, and the 2026 counterexamples do not settle the related Dixmier conjecture for the first Weyl algebra [Off]. [background: the general conjecture is stably equivalent to the Dixmier conjecture [BKK]]

9 Proved and open

Statement	Status	Source
JC_1	true	elementary [MY, Thm. 4.1(i)]
Keller maps of degree ≤ 2, any n	true	Wang 1980 [Wang]
JC_2 for degrees ≤ 100	true	Moh 1983 [Moh]
JC_3	**false**	explicit map (1) [Alp]; Isabelle [AFP], Lean [SM, NG, XL]
JC_n, every $n \geq 3$	**false**	padding [AFP, p. 18]
MathDB's formal statement (every $n \geq 1$)	**false**	Lean refutation [XL]
JC_2 in general	**open**	the problem on MathDB [Off]

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